M. Masood ul Rahman Usmani

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About

I am a highly motivated professional with a Master of Science degree in Computer Science from COMSATS University Islamabad, Sahiwal campus. My expertise lies in the areas of the Machine Learning, Deep Learning, C#, OOP, Java, Python, and C++. With over 1 years of experience as a Computer Lecturer in Lodhran, I have developed a strong skillset in teaching courses such as Data Science, Android App Development, cloud computing, and Machine Learning.
As a Project Supervisor in different academic institutions for MCS, BS, and ADP, I have gained valuable experience in managing projects and mentoring students. I am a creative problem-solver and a collaborative team player who is committed to delivering high-quality work.

Experience

Visiting Lecturer — Sep 2024 – ongoing
Bahauddin Zakariya University, Lodhran Sub Campus — *Lodhran, Pakistan*

- Facilitated learning for undergraduate students through engaging and informative lectures.
- Provided comprehensive evaluation and maintained accurate records for students, including grading classwork, laboratory work, assignments, papers, attendance, and other necessary data.

Visiting Lecturer — Oct 2025 – Jan 2026
The Islamia University of Bahawalpur — *Bahawalpur, Pakistan*

- Facilitated learning for undergraduate students through engaging and informative lectures.
- Provided comprehensive evaluation and maintained accurate records for students, including grading classwork, laboratory work, assignments, papers, attendance, and other necessary data.

Visitng Lecturer — sep 2023 – May 2025
Punjab Group of Colleges — *Lodhran, Pakistan*

- Developed and delivered engaging lectures for undergraduate BS program in **Cloud Computing and Data Science**.
- Led interactive lab sessions, providing practical experience with Machine Learning technologies

College Teaching Intern — Nov 2024 – April 2025
Govt. Graduate college, Lodhran — *Lodhran, Pakistan*

- Facilitated learning for undergraduate students through engaging and informative lectures.
- Provided comprehensive evaluation and maintained accurate records for students, including grading classwork, laboratory work, assignments, papers, attendance, and other necessary data.

Publications

1. M. Rashid, **M. M. ul Rehman**, H. Raza, M. Qadeer, I. Rasool, and N. A. Zafar, "Formal Modeling and Verification of IoT-Based Smart Waste Management System Using SPIN".
2. M. Rashid, M. Qadeer, H. Raza, **M. M. U. Rehman**, I. Rasool, and N. A. Zafar, "Verification of Safety of Aircraft Arrival Procedure Using SPIN Model Checker," in Proc. 2023 Int. Conf. Frontiers Inf. Technol. (FIT), Dec. 2023, pp. 262, 267.
3. M. Rashid, **M. M. ul Rehman**, H. Raza, M. Qadeer, I. Rasool, and N. A. Zafar, "Formal Verification of Aircraft Departure Procedure Using SPIN Model Checker," in Proc. 2023 18th Int. Conf. Emerging Technol. (ICET), Nov. 2023, pp. 165, 170.
4. Makki Riaz Khan, **M. M. U. R. Usmani**, Muzamil Mehboob, Faheem Abbas, Shahnaz Rafique, and Kishwar Bibi, "Multi-Class Classification of Alzheimer's Impairment and Dementia Stages Using an EfficientNet-B0 Deep Learning Framework," *Journal of Computing and Biomedical Informatics*, vol. 11, no. 01, 2026.
5. **M. M. U. R. Usmani**, R. Rafiq, and M. Farhan, "An ML Enabled Internet of Things Framework for Diabetes Management," *The Journal of Supercomputing*, Under Revision.
6. **M. M. U. R. Usmani** and M. R. Khan, "Pneumonia Classification Using DenseNet201 on Custom Combined Dataset," *Journal of AI and Data Mining*, Submitted.
7. **M. M. U. R. Usmani**, Rimsha Rafiq, and Makki Riaz Khan, "IoT Enabled Deep Reinforcement Learning for Adaptive Waste Management in Hospital Environments," *Latin American Journal of Computing*, Under Review (2nd Phase).
8. Makki Riaz Khan, **M. M. U. R. Usmani**, "Improving GDP Growth and Inflation Forecasting Using a Hybrid ARIMA and Deep Learning Approach," *KIET Journal of Computing and Information Sciences*, Submitted.

Academic Projects

COVID Detection from Chest X-Ray images using Convolutional Neural Network

* Created a Convolutional Neural Network (CNN) model to detect COVID-19 from chest X-ray images. Integrated with an HTML interface, allowing users to upload images, process them, and display results on the same page for quick and easy diagnosis.

School Management System

* Developed a School Management Software using HTML, CSS, jQuery, PHP, and MySQL to streamline administrative operations and track student information efficiently.

Pneumonia Detection from Chest X-Ray images using Convolutional Neural Network

* Developed a Convolutional Neural Network (CNN) model to detect pneumonia from chest X-ray images. Integrated the model with an HTML interface for direct image uploads, processing, and result display, enabling quick and easy diagnosis.

Heart Failure Prediction using Machine Learning

* Predictive Modeling: The project leverages machine learning algorithms to predict heart failure events based on patient data, including clinical and demographic variables, to enhance early diagnosis and treatment strategies.
* Data-Driven Insights: Utilizes a comprehensive dataset containing various patient health metrics, aiming to identify key factors contributing to heart failure, thereby offering valuable insights for healthcare professionals.

Live Face Detection using open CV

* Real-time Face Detection: The project implements a live face detection system using OpenCV and Python, enabling real-time identification of faces through a webcam or video feed.
* Efficient and Lightweight: The code is optimized for efficient performance, making it suitable for use on standard computing devices without requiring specialized hardware.

Lungs cancer prediction using Machine Learning

* Project Overview: This project focuses on predicting lung cancer using machine learning algorithms. It involves the implementation of various classification techniques to analyze and predict the likelihood of lung cancer based on patient data.
* Technical Implementation: The project is developed in Python, utilizing libraries such as Scikit-learn, Pandas, and NumPy for data preprocessing, feature selection, and model training. The dataset undergoes rigorous cleaning and transformation to ensure accurate predictions.

SONAR Rock vs Mine Prediction Using machine Learning Techniques

* Objective: The project aims to classify objects as either rocks or mines using sonar signal data by implementing machine learning algorithms.
* Dataset: Utilizes the UCI Machine Learning Repository's Sonar dataset, which contains 208 instances of sonar signals reflected off rocks and mines, with 60 features representing signal strength at various angles.
* Modeling: Implements various machine learning models, including Logistic Regression, Support Vector Machine (SVM), and Random Forest, to achieve accurate predictions with performance metrics such as accuracy, precision, and recall evaluated.

Certifications

- AI for Beginners - HP Life
- Introduction to CyberSecurity Awareness - Hp Life
- Data Science & Analytics - HP Life
- Design Patterns made-easy - Udemy
- Refactoring Techniques made-easy - Udemy
- Deep Reinforcement Learning made-easy - Udemy
- Code Smells made-easy - Udemy
- Specialization Certificate In Machine Learning - Coursera
- Supervised Machine Learning: Regression and Classification - Coursera
- Advanced Learning Algorithms - Coursera
- Unsupervised Learning, Recommenders, Reinforcement Learning - Coursera
- Intro to Programming - Kaggle
- Python - Kaggle
- Intro to Machine Learning - Kaggle
- pandas - Kaggle
- Intermediate Machine Learning - Kaggle
- Feature Engineering - Kaggle
- Intro to SQL - Kaggle
- Advance SQL - Kaggle
- Intro to Deep Learning - Kaggle
- Computer Vision - Kaggle
- Data Cleaning - Kaggle
- Intro to AI Ethics - Kaggle
- Machine Learning Explainability - Kaggle
- Intro to Game AI and Reinforcement Learning - Kaggle
- Certificate of participation - International Conference on Frontiers of Information Technology 2023

EDUCATION

The Islamia University of Bahawalpur — Bahawalpur, Pakistan
Bachelor of Science(B.Sc.) Marks: 519/800 — *2018 – 2020*

The Islamia University of Bahawalpur — Bahawalpur, Pakistan
Master of Computer Science CGPA: 3.21/4 — *Sep 2020 – Feb 2022*

COMSATS University Islamabad — Sahiwal, Pakistan
MS Computer Science CGPA: 3.87/4 — *2022 – July 2024*
Thesis: An ML-Enabled Internet of Things Framework for Diabetes management

TECHNICAL SKILLS

- Proficient coder in multiple languages and environments including:
  1. Python
  2. Scikit-learn
  3. SciPy
  4. TensorFlow
  5. Keras
  6. PyTorch
  7. OpenAI Gym
  8. Matplotlib
  9. Seaborn
  10. OpenCV
  11. C
  12. C++
  13. C#
  14. Java (Android)
  15. OOP
- Experienced in developing AI-based applications using:
  * Machine Learning
  * Deep Learning
  * Reinforcement Learning